ASSIGNMENT - 7.5

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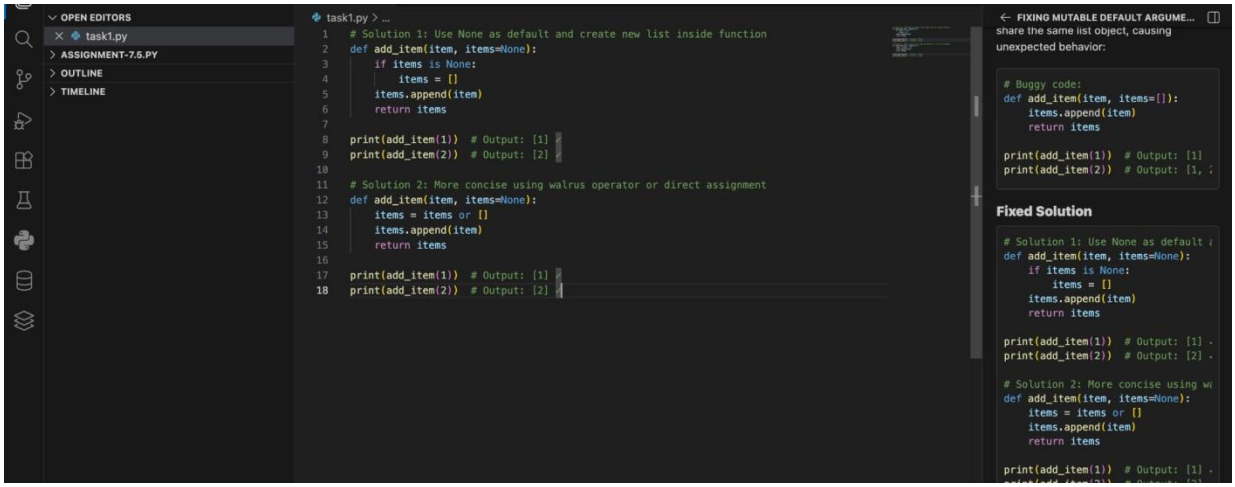
Batch:29

Task1(MutableDefaultArgument–FunctionBug)

Task: Analyze given code where a mutable default argument causes unexpected behavior. Use AI to fix it.

```
#Bug:Mutabledefaultargument def
add_item(item, items=[]):
    items.append(item)
    return items
print(add_item(1))
print(add_item(2))
```

Expected Output: Corrected function avoids shared list bug.



Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                   Python

arepallyramcharan@Arepallys-MacBook-Air assignment-7.5.py % /usr/bin/python3 "/Users/arepallyramcharan/ai assist
ent.py/assignment-7.5.py/task1.py"
[1]
[2]
[1]
[2]
arepallyramcharan@Arepallys-MacBook-Air assignment-7.5.py %
```

Observation:

Defaultmutablearguments(likelists)arecreatedonlyonceatfunctiondefinition time.

The same list is reused across multiple function calls.

Thiscausesunintendeddatasharingandunexpectedbehavior.

Using `None`as a default value prevents shared state issues.

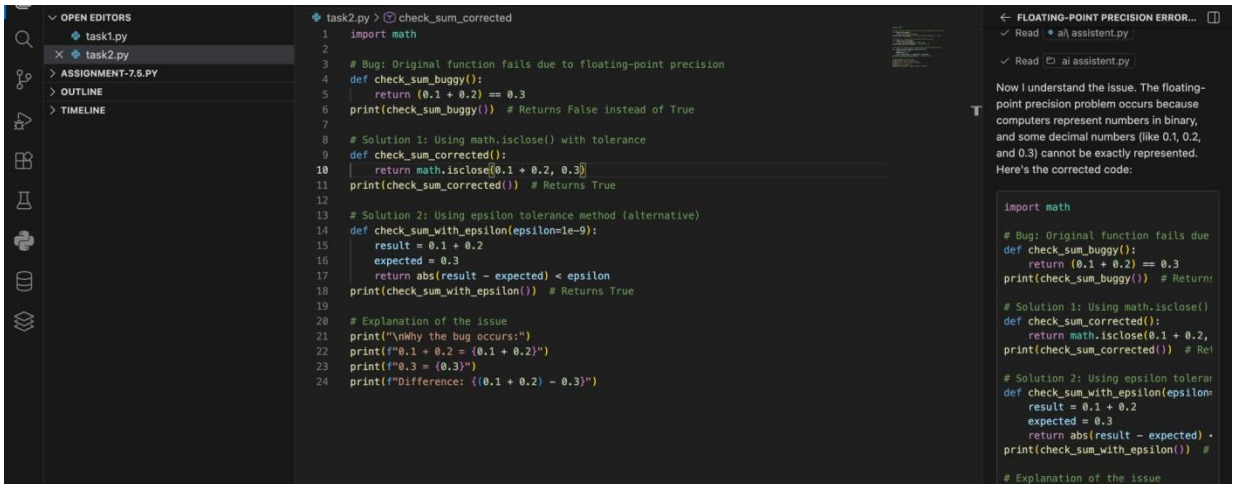
Properhandling of default arguments improves code reliability and predictability.

Task2(Floating-PointPrecisionError)

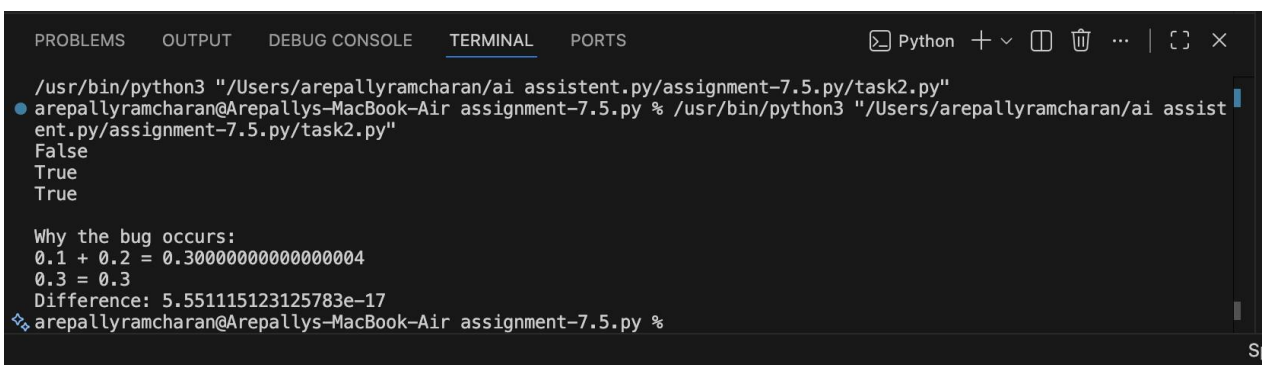
Task: Analyze given code where floating-point comparison fails. Use AI to correct with tolerance.

```
#Bug:Floatingpointprecisionissue def
check_sum():
    return(0.1+0.2)==0.3
print(check_sum())
```

Expected Output: Corrected function



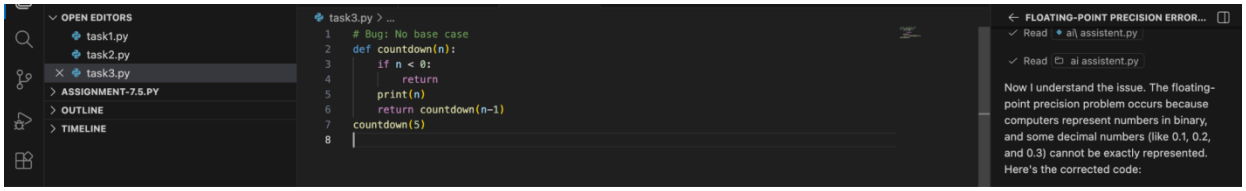
Output:



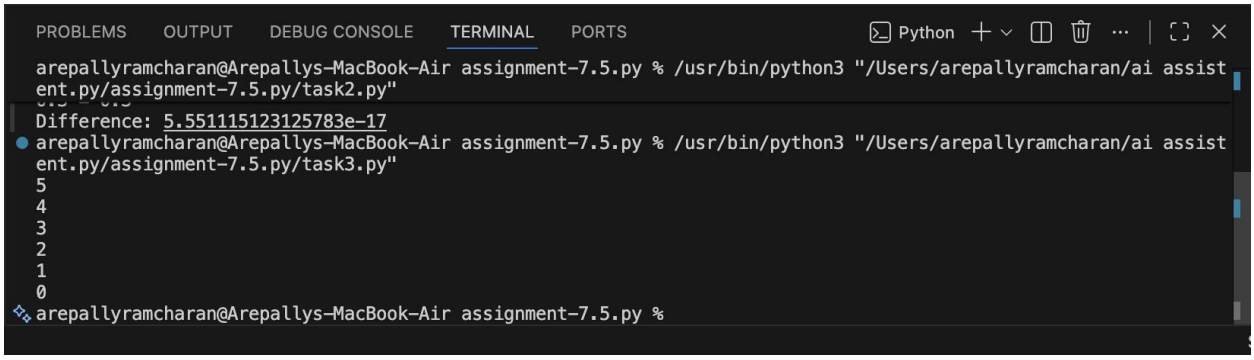
Observation:

- Floating-pointnumbers are not always stored exactly in memory.
- Direct equality comparison (`==`) may fail due to small precision errors.
- Using a tolerance value avoids strict comparison issues.
- `math.isclose()` is the recommended safe comparison method.
- Properfloating-point handling improves numerical accuracy and reliability.

TaskDescription#3(TransparencyinErrorHandling)
Task:UseAItogenerateaPythonprogramthatreadsafileand processes data.
Prompt:
“Generatecodewithpropererrorhandlingandclearexplanations for each exception.”
Expected Output:
• Code with meaningful exception handling.
• Clear comments explaining each error scenario.
• Validationthatexplanationsalignwithruntimebehavior.



Output:



Observation:

1. Recursion must always include a base case to stop execution.
2. Without a stopping condition, infinite recursion occurs.
3. Python limits recursion depth to prevent crashes.
4. Each recursive call should move toward the base case.
5. Proper base case design ensures predictable and safe recursion.

Task4(DictionaryKeyError)

Task:Analyzegivencodewhereamissingdictionarykeycauseserror.UseAItofixit. # Bug: Accessing non-existing key

```
def get_value():
    data={"a":1,"b":2}
    return data["c"]
print(get_value())
```

Expected Output: Corrected with .get() or error handling.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                    Python

/usr/bin/python3 "/Users/arepallyramcharan/ai assistent.py/assignment-7.5.py/task4.py"
arepallyramcharan@Arepallys-MacBook-Air assignment-7.5.py % /usr/bin/python3 "/Users/arepallyramcharan/ai assist
ent.py/assignment-7.5.py/task4.py"
None
arepallyramcharan@Arepallys-MacBook-Air assignment-7.5.py %
```

Observation:

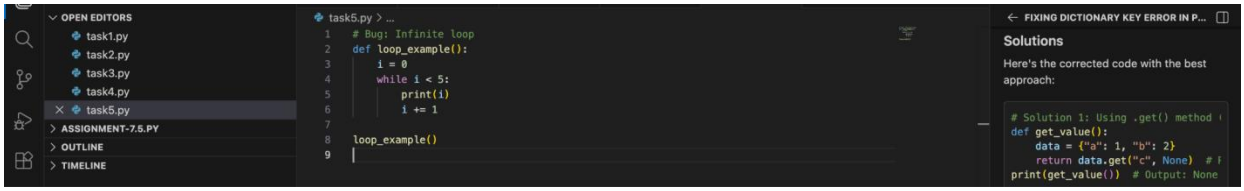
1. Direct dictionary access raises `KeyError` if the key does not exist.
2. `.get()` method provides safer access with a default value.
3. Exception handling prevents program crashes.
4. Defensive programming improves reliability.
5. Proper key validation ensures stable execution.

Task5(InfiniteLoop–WrongCondition)

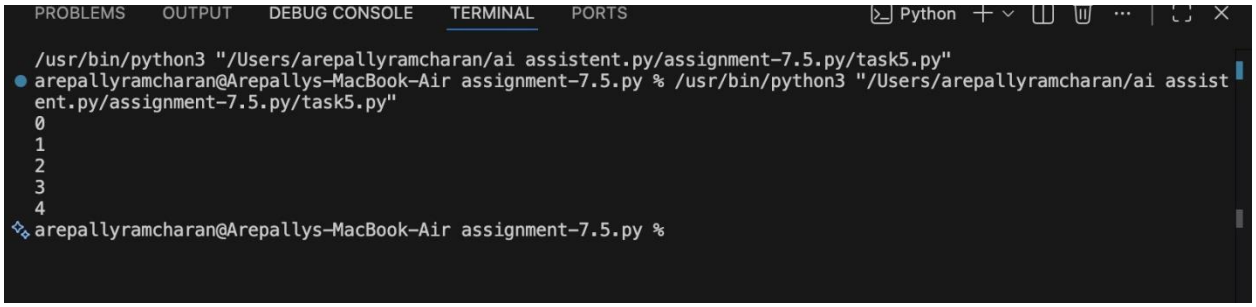
Task:Analyzegivencodewhereloopneverends.UseAItodetectandfixit. # Bug: Infinite loop

```
defloop_example():
    i = 0
    whilei<5: print(i)
```

Expected Output: Corrected loop increments i.



Output:



Observation:

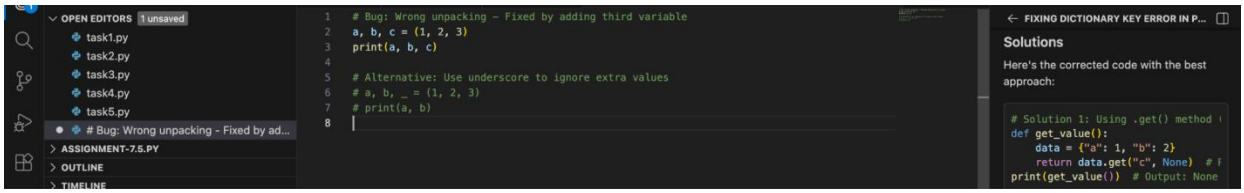
- Infinite loops occur when the loop variable never changes.
- The stopping condition must eventually become False.
- Loop control variables must be updated inside the loop.
- Careful condition checking prevents runtime hangs.
- Properloop design ensures predictable execution flow.

Task6(UnpackingError–WrongVariables)

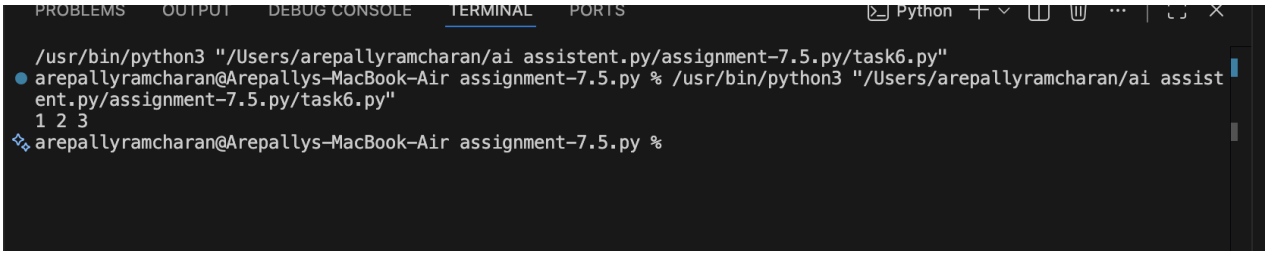
Task:Analyzegivencodewheretupleunpackingfails.UseAItofixit. # Bug: Wrong unpacking

a, b = (1, 2, 3)

Expected Output: Correct unpacking or using _ for extra values.



Output:



Observation:

1. Number of variables must match number of elements in unpacking.
2. Mismatch causes `ValueError`.
3. `_` can be used to ignore unwanted values.
4. `*` operator allows flexible unpacking of remaining values.
5. Proper unpacking improves code readability and prevents runtime errors.

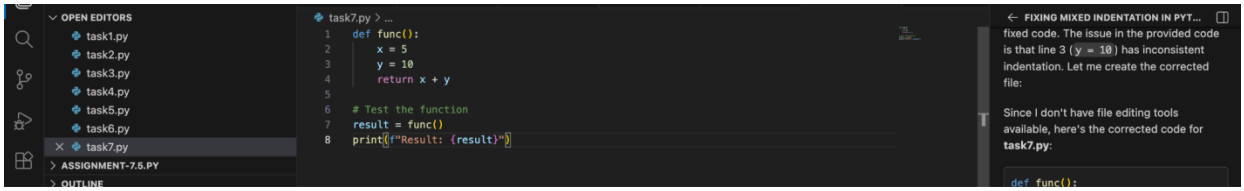
Task7(MixedIndentation–TabsvsSpaces)

Task: Analyzegivencodewheremixedindentationbreaksexecution.UseAItofixit. # Bug: Mixed indentation

```
deffunc():
    x = 5
        y = 10
```

return x+y

Expected Output : Consistent indentation applied.



Output:

Observation:

1. Python strictly enforces indentation rules.
2. Mixingtabs and spaces leads to `IndentationError`or `TabError`.
3. Allstatements inside a block must align consistently.
4. Standard convention is 4 spaces per indentation level.
5. Proper indentation improves readability and prevents syntax errors.

Task8(ImportError–WrongModuleUsage)

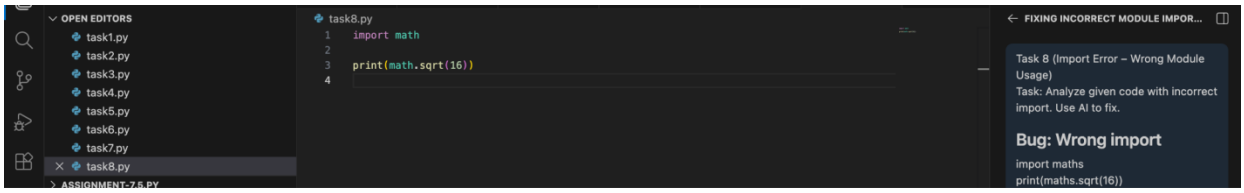
Task:Analyzegivencodewithincorrectimport.UseAItofix. #

Bug: Wrong import

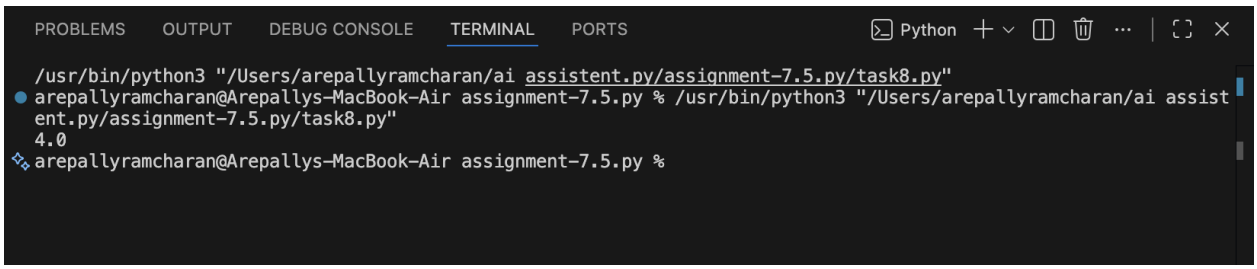
import maths

print(maths.sqrt(16))

Expected Output: Corrected to import math



Output:



Observation:

1. Python module names must be spelled correctly.
2. Incorrectmodule names cause `ModuleNotFoundError`.
3. Python module names are case-sensitive.
4. Always verify module documentation before importing.
5. Correct imports ensure proper access to built-in functions.

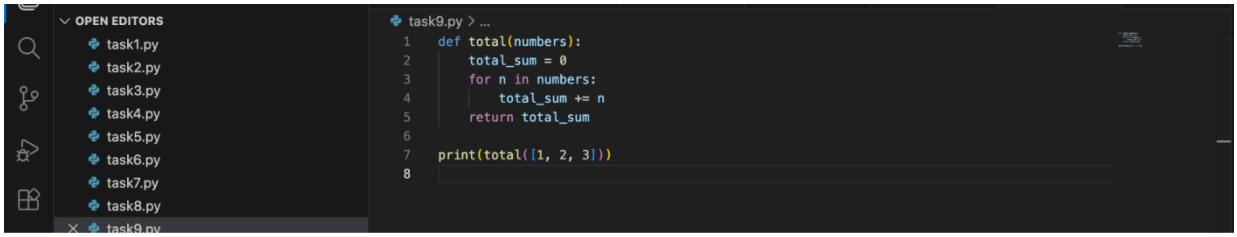
Task9(UnreachableCode–ReturnInsideLoop)

Task: Analyzegivencodewhereareturninsidealoppreventsfulliteration.UseAItofixit. # Bug: Early return inside loop

```
def total(numbers):
    for n in numbers:
        return n
```

print(total([1,2,3]))

Expected Output: Corrected code accumulates sum and returns after loop.



Output:

Observation:

1. A `return`statement immediately exits the function.
2. Placing `return`inside a loop may stop full iteration.
3. Code after early return becomes unreachable.
4. Accumulation logic should complete before returning result.
5. Proper return placement ensures correct computation.

Task10(NameError–UndefinedVariable)

Task:Analyzegivencodewhereavariableisusedbeforebeingdefined.LetAIdetectandfix the error.

Bug: Using undefined variable

```
def calculate_area():
    return length * width
print(calculate_area())
```

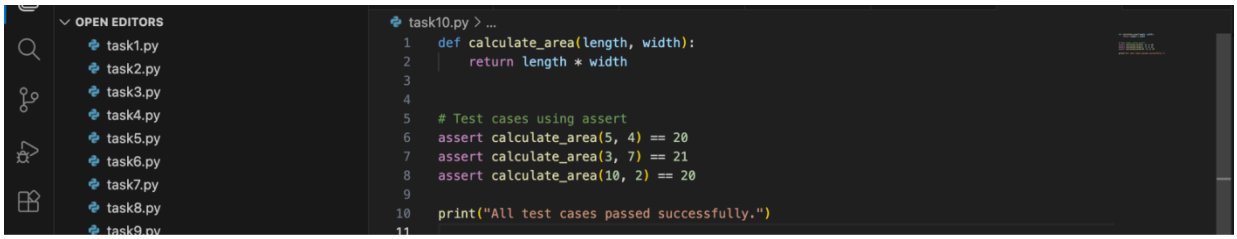
Requirements:

- Run the code to observe the error.
- Ask AI to identify the missing variable definition.
- Fix the bug by defining length and width as parameters.
- Add 3 assert test cases for correctness.

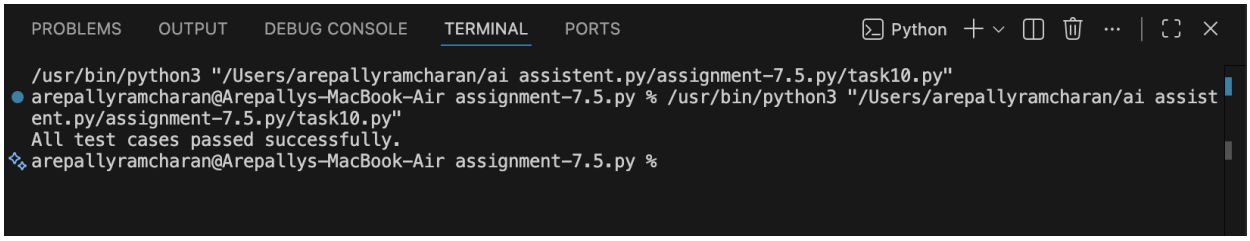
Expected Output :

- Corrected code with parameters.
- AI explanation of the bug.

Successful execution of assertions.



Output:



Observation:

1. The function attempted to use variables that were not defined.

2. Variablesinsideafunctionmusteither:
    - Be defined inside the function, or
    - Be passed as parameters.
3. Since`length`and `width`were missing, Python raised a `NameError`.
4. Adding them as function parameters fixes the scope issue.
5. Assertions confirm correctness of the function logic.

Task11(TypeError–MixingDataTypesIncorrectly)

Task:Analyzegivencodewhereintegersandstringsareaddedincorrectly.LetAIdetectandfix the error.

#Bug:Addingintegerandstring def
add_values():
    return 5 +
"10"print(add_values())

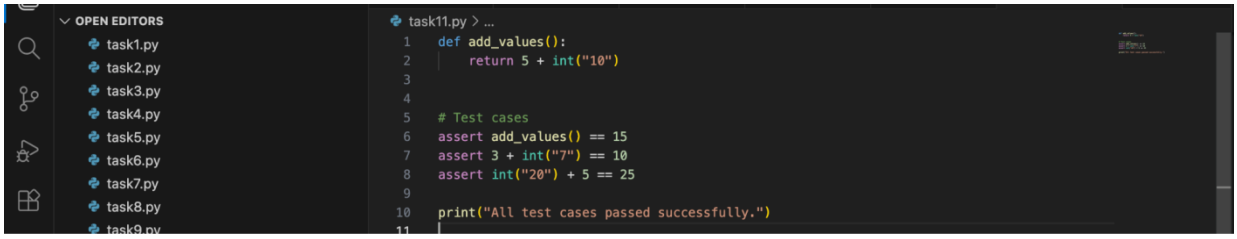
Requirements:

- Run the code to observe the error.
- AI should explain why int + str is invalid.
- Fix the code by type conversion (e.g., int("10") or str(5)).
- Verifywith3assertcases.

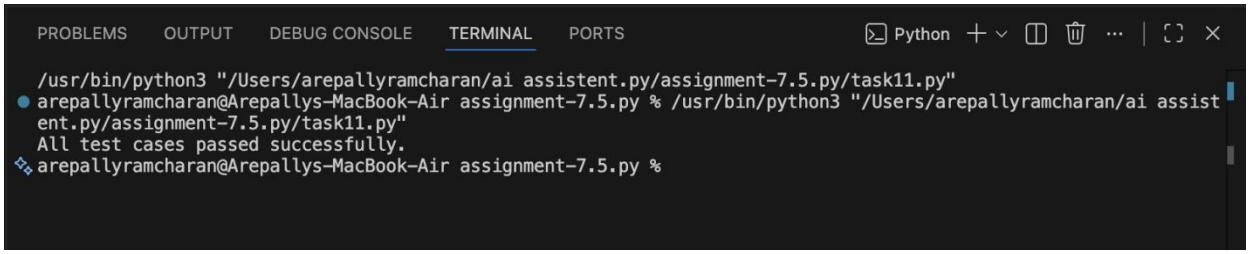
Expected Output #6:

- Corrected code with type handling.
- AI explanation of the fix.

Successful test validation.



Output:



Observation:

1. The error occurs because Python is strongly typed.
2. Integers and strings cannot be added directly.
3. Explicit type conversion is required.
4. Converting `"10"`to `int`enables numeric addition.
5. Assertionsconfirm the corrected logic works properly.

Task12(TypeError–String+ListConcatenation)

Task:Analyzecodewhereastringisincorrectlyaddedtoalist. #
Bug: Adding string and list

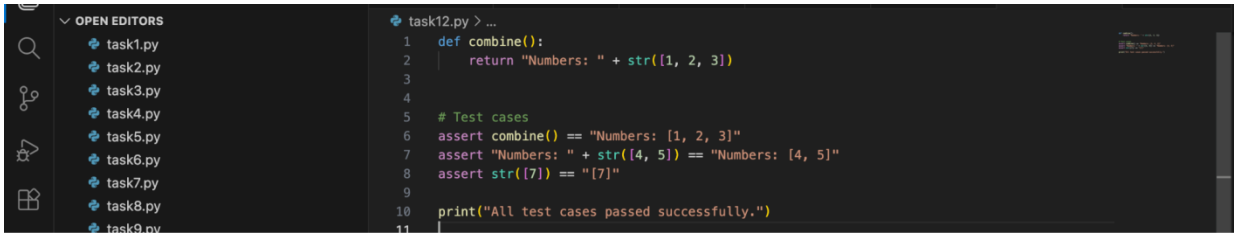
```
def combine():
    return"Numbers:"+[1,2,3]
print(combine())
```

Requirements:

- Run the code to observe the error.
- Explain why str + list is invalid.
- Fix using conversion (str([1,2,3]) or "".join()).
- Verifywith3assertcases.

Expected Output:

- Corrected code
- Explanation
- Successful test validation



Output:

Observation:

1. Python is strongly typed and does not allow `str + list`.
2. Both operands must be of the same type for concatenation.
3. `str([1,2,3])` converts the list to a string representation.
4. `"".join(map(str,list))` convertseachelementtostringandjoinsthem cleanly.
5. Assertionsconfirm the corrected function works properly.

Task13(TypeError–MultiplyingStringbyFloat)

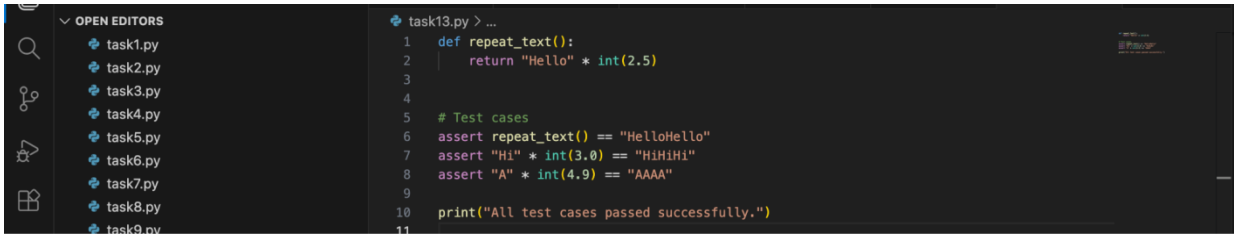
Task:Detectandfixcodewhereastringismultipliedbyafloat. # Bug:

Multiplying string by float

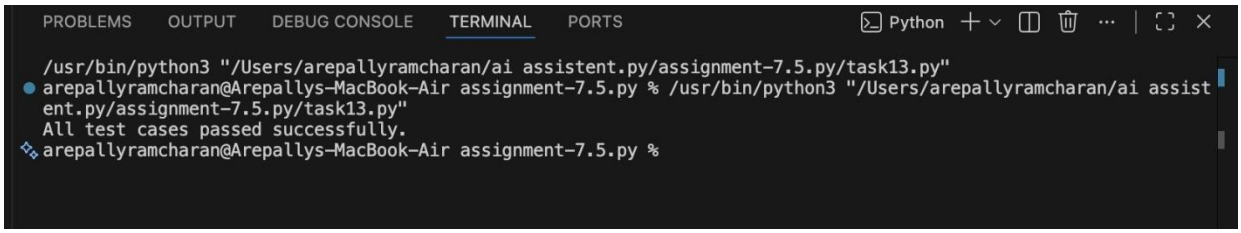
```
def repeat_text():
  return"Hello"*2.5
print(repeat_text())
```

Requirements:

- Observe the error.
- Explain why float multiplication is invalid for strings.
- Fix by converting float to int.
- Add 3 assert test cases.



Output:



Observation:

String multiplication requires an integer.

Floatvaluesmustbeconvertedusing`int()`.

`int(2.5)` becomes `2`(truncates decimal).

Afterconversion,repetitionworkscorrectly.

Assertions verify correct functionality.

Task 14 (Type Error – Adding None to Integer)

Task:AnalyzecodewhereNoneisaddedtoaninteger. #

Bug: Adding None and integer

def compute():

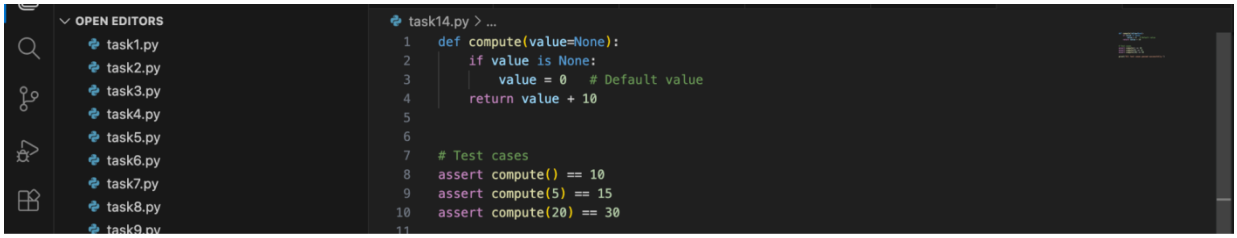
value = None

returnvalue+10

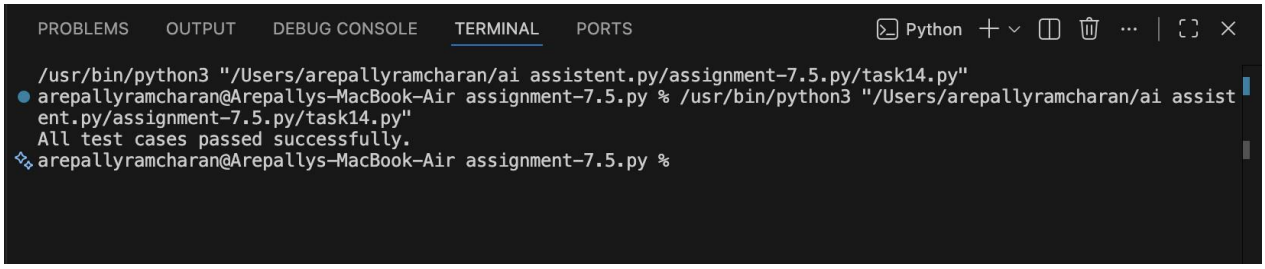
print(compute())

Requirements:

- Run and identify the error.
- ExplainwhyNoneTypecannotbeadded.
- Fix by assigning a default value.
- Validateusingasserts.



Output:



Observation:

`None`cannotparticipateinarithmeticoperations. We

check if `value is None`.

Ifso,assignasafedefaultnumericvalue(`0`). This

prevents TypeError.

Assertions confirm correct behavior.

Task15(TypeError–InputTreatedasStringInsteadofNumber)

Task:Fixcodewhereuserinputisnotconvertedproperly. #

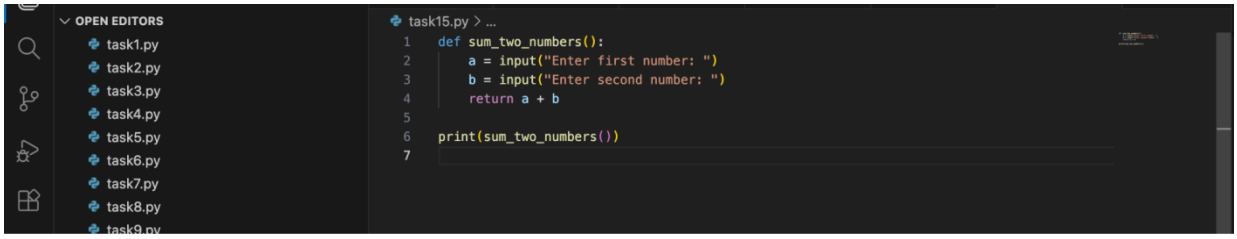
Bug: Input remains string

```
def sum_two_numbers():
    a = input("Enter first number: ")
    b=input("Entersecondnumber:")
    return a + b

print(sum_two_numbers())
```

Requirements:

- Explain why input is always string.
- Fix using int() conversion.
- Verifywithasserttestcases.



Output:

Observation:

`input()` always returns string data.

Stringadditionresultsinconcatenation. Use

`int()` to convert string to integer.

Afterconversion,`+`performsnumericaddition.

Assertions confirm correct functionality.